Resolution Geometry: Tensor Frontier

Forecast Horizons, Tidal Curvature, the Source and the Observer's Boundary

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November 13, 2026 (second edition)

Abstract

A forecast made from a record of precision σ_0 stays within a tolerance ε, with coverage q, up to a horizon that grows like $\ln(1/\sigma_0)/\lambda$. This print asks what that law is in Resolution Geometry and what it does for the tensor. Six results come out.

The frontier is already in the ladder. The predictive metric of the Tensor Derivation is the generating function of the forecast-error profile. The exponent λ is its critical discount: the predictive metric is Lipschitz-equivalent to the present metric for $\beta < e^{-\lambda}$ and only Hölder, with exponent $\ln(1/\beta)/\lambda$, above.

It is an upper bound. For a Gaussian record no forecaster exceeds the coverage of the posterior-mean forecast, so the horizon bounds every method that uses that record. An unrecorded sector caps it at a value that no improvement of the recorded precision moves. For fluctuating growth the horizon is $L/\lambda - z_q\sqrt{DL/\lambda^3} + O(1)$ with $L = \ln(\varepsilon/\sigma_0)$; the logistic map at $r = 4$ has $D = 0$ and cannot show the second term.

Signature. On a lattice of spacing a the horizon of a field with principal form of signature (p, q) is proportional to a, unless $\min(p, q) = 1$. A free Lorentzian field has no frontier at all, since isometries of a positive form do not grow. For the interacting quartic field in $3 + 1$ dimensions the growth rate is bounded independently of a.

Tensor. For two free carriers at relative rest the forecast propagator obeys $\ddot{\Phi} = -E\Phi$ with the tidal tensor $E_{ij} = R_{i0j0}$. Hence $E = -\ddot{\Phi}(0)$; the frontier exponents are $\sqrt{-e_i}$ for the negative eigenvalues of E; and the Fisher volume of the initial separation obeys $\partial_t^2 \ln\det I_t|_0 = -2\,\mathrm{Ric}(u, u)$. The response law of the Tensor Derivation holds at a point exactly when this volume law holds there for all observers with the source on the right-hand side. Ten generic observers recover the source; observers with collinear or coplanar velocities see three or six of its ten components, and the rest is in their hole.

Sources are moments. Coverage does not control a quadratic source: the mean square error is at most $\varepsilon^2 + \sqrt{(1-q)m_4}$, and this is sharp. The expected source exceeds the source of the forecast by a positive defect that grows at twice the exponent.

The frontier is the observer's boundary. A record covers a bounded region. For dynamics with a cone of speed C the exterior is exactly closed off up to the crossing time, and for a massless carrier the horizon of a record of radius R is R/C at every precision: $C = \mathrm{d}R/\mathrm{d}H$. Mass and dispersion move the bulk and threshold arrivals, not the front. The frontier is at most the invariant speed K, and a carrier whose two-way speed is the same for all observers attains it. An audit of the supplied gates finds that they would lock the speed of sound, that their lock depends on the seed (106 of 200), and that one gate needs the answer; two gates are added, boost and maximality. On published raw data the speed of light is 53 835.1286(2) krypton wavelengths per caesium period, and the gates hold at 2×10^{-18} (isotropy), 2×10^{-13} (observer velocity), 2×10^{-9} (source velocity) and 3×10^{-15} (gravitational against electromagnetic carrier). That is consistency at those bounds, not a proof.

One tensor frontier has a published number. The damping rate of black-hole ringdown tends to half the instability exponent of the light ring; computed quasinormal frequencies show the ratio 0.9245 at $l = 2$ and 0.9974 at $l = 12$, and for the remnant of GW150914 twice

the light-ring e-folding time is 3.80 ms against a published 4.0 ± 0.3 ms. A battery of 17 groups, each with a canary, checks every statement; 22 mutations are caught. Nothing here derives the response law, and Gate A, the nonlinear completion and the gauge sectors are untouched.

Contents

1 Purpose, verdict and status **2**

2 The frontier as an object of Resolution Geometry **4**
2.1 The predictive metric is its generating function 4
2.2 The frontier is an upper bound 4
2.3 The second term 5
2.4 The hole caps the horizon 5

3 Refinement, signature and the free field **6**

4 The tensor **7**
4.1 The slope does not depend on the record 7
4.2 Curvature is the generator of the forecast 7
4.3 The source is the rate of the Fisher volume 8

5 The light ring: a tensor frontier with a published number **9**

6 Sources are moments **10**

7 The frontier is the observer's boundary **11**
7.1 The causal cap 11
7.2 Front, group and threshold 11
7.3 Maximal means invariant 12

8 The raw speed: gates and data **13**
8.1 Audit of the supplied gates 13
8.2 Published raw data 13

9 What is closed and what is not **14**

10 Predictions and falsifiers **15**

11 Limits **16**

A Battery **17**

1 Purpose, verdict and status

The input is a forecast law, an executed set of 29 checks on it, and a conjecture: that the law closes open points of the tensor programme or extends its derivation. The conjecture is correct in part. This section says which part.

Definition 1.1 (Forecast task)**.** A forecast task consists of a record of the initial state with error scale σ_0, an update U, a later record T with tolerance ε, and a coverage $q < 1$. The *horizon* n_q is the largest time up to which the forecast error stays within ε with probability at least q. The *frontier* is the function $\sigma_0 \mapsto n_q$.

For a single unstable direction with constant growth $e^{\lambda n}$ and initial error $\sigma_0 Z$,

$$n_q = \frac{\ln(\varepsilon/\sigma_0) - \ln Q_q(|Z|)}{\lambda}, \tag{1}$$

up to the integer part. The slope in $\ln(1/\sigma_0)$ is $1/\lambda$; the law of the initial error enters only through the intercept. The supplied checks establish this under a fixed task, coverage and protocol family, and mark where broader readings fail. They are taken as given. What follows is new relative to them.

Point of the tensor programme	Effect	Where
Predictive metric (Tensor Derivation)	EXTENDED	generating function; critical discount (Theorem 2.1)
Signature from predictive closure	SHARPENED	horizon proportional to the spacing (Theorem 3.1)
Free and interacting Lorentzian fields	CLOSED	no frontier; uniform bound (Theorem 3.2, Proposition 3.4)
Curvature as a record	CONSTRUCTED	$E = -\ddot{\Phi}(0)$; exponents (Theorem 4.2)
Source as a record	CONSTRUCTED	volume law; equivalence with the response law (Theorems 4.3, 4.4)
Shape of the source, ten components	CONSTRUCTED	observer rank $1, 3, 6, 10$ (Theorem 4.4)
Energy deficit (GR Automation)	EXTENDED	forecast deficit, twice the exponent (Theorem 6.2)
Gate R2, sources as first variations	EXTENDED	transfer inequality; concentration defect (Theorem 6.1)
Cone and forecast (order and cone rungs)	CONSTRUCTED	causal cap $H = R/C$ (Theorem 7.1)
Frontier and the speed of light	CONSISTENT	invariant speed; nine gates on published data (Theorem 7.3, Section 8)
Gate A; nonlinear completion; inhomogeneous stability; gauge sectors	UNTOUCHED	—

Three corrections to the supplied analysis were found on the way.

- Its nonlinear run uses the logistic map at $r = 4$. That map is a coboundary (Theorem 2.3), so its finite-time exponent does not fluctuate at order $\sqrt{n}$. The tidy constant intercept seen there is a property of that map.
- The growing correction it found for a random cocycle is not a qualitative remark. Its coefficient is $-z_q\sqrt{D/\lambda^3}$.
- The law was tested for one forecaster. It bounds all of them (Theorem 2.2).

A second supplied programme, a synthetic test of a raw speed estimator with seven gates, is audited in Section 8; it prompted Sections 7 and 8 of this edition.

Status words: PROVED here; IDENTIFIED with a known result, which is named; IMPORTED; COMPUTED by the battery. Battery groups are quoted in brackets.

2 The frontier as an object of Resolution Geometry

2.1 The predictive metric is its generating function

The Tensor Derivation compares states by the predictive metric

$$d_\beta(x,y) = (1-\beta)\sum_{n\geq 0}\beta^n d_0(U^n x, U^n y), \tag{2}$$

with $d_0 \leq 1$ the present-observation distance and $0<\beta<1$. Write $e_n = d_0(U^n x, U^n y)$ for the *error profile* of the pair, and $n_\varepsilon = \min\{n : e_n > \varepsilon\}$ for its horizon.

Theorem 2.1 (Generating function and critical discount)**.** [FA]

1. *$d_\beta/(1-\beta)$ is the generating function of the error profile. The family $\{d_\beta\}_{0<\beta<1}$ determines the profile, and with it every horizon.*

2. *For every profile and every ε,*

$$(1-\beta)\,\varepsilon\,\beta^{n_\varepsilon} \;\leq\; d_\beta \;\leq\; \varepsilon + \beta^{n_\varepsilon}, \tag{3}$$

   with $\beta^\infty = 0$ when the tolerance is never exceeded.

3. *If $e_n = \min(1, \delta e^{\lambda n})$, then $d_\beta \asymp \delta^\alpha$ as $\delta \to 0$, with*

$$\alpha = \min\Big(1, \frac{\ln(1/\beta)}{\lambda}\Big), \tag{4}$$

   and a factor $\ln(1/\delta)$ at $\beta = e^{-\lambda}$.

Proof. (1) A power series is determined by its values on an interval. (2) The term n_ε alone exceeds the lower bound. For the upper bound, the terms before n_ε sum to at most $\varepsilon(1-\beta^{n_\varepsilon})$ and the others to at most β^{n_ε}. (3) Let $n^* = \ln(1/\delta)/\lambda$. The saturated tail contributes $\beta^{n^*} = \delta^{\ln(1/\beta)/\lambda}$. The growing part is a geometric sum with ratio βe^λ: for $\beta e^\lambda > 1$ it is comparable to $\delta(\beta e^\lambda)^{n^*} = \beta^{n^*}$, and for $\beta e^\lambda < 1$ to δ, which then dominates. □

The Tensor Derivation shows that U is β^{-1}-Lipschitz for d_β, for every dynamics and every β. Clause (3) adds what that statement cannot see. The identity map from the present metric to the predictive metric is bi-Lipschitz below the critical discount $\beta_c = e^{-\lambda}$ and a snowflake above it. The exponent is thus read from records alone, as the discount at which the predictive metric stops being equivalent to the present one. For the doubling map with $\delta = 2^{-k}/3$ the profile is exactly $\min(2^{n-k}, 1)/3$; the measured exponents are $1, 1, 0.7370, 0.4150, 0.1520$ at $\beta = 0.3, 0.4, 0.6, 0.75, 0.9$, equal to the predicted ones to 10^{-7}.

2.2 The frontier is an upper bound

Theorem 2.2 (No forecaster beats the frontier)**.** [FB] *Let the target T_n be Gaussian given the record, with standard deviation s_n not depending on the value of the record. Then every forecaster $\hat{t}$, any measurable function of the record, satisfies*

$$\Pr(|T_n - \hat{t}| \leq \varepsilon) \leq 2\Phi(\varepsilon/s_n) - 1, \tag{5}$$

with equality for the posterior mean. If $s_n = s_0 e^{\lambda n}$, no method has a horizon beyond $[\ln(\varepsilon/s_0) - \ln z_{(1+q)/2}]/\lambda$.

Proof. Condition on the record. An interval of length 2ε carries the largest Gaussian mass when it is centred at the mean. For vector targets and convex symmetric tolerance sets this is Anderson's lemma [1]. □

The battery takes a prior of unit scale, a record of precision 10^{-4} and $n = 9$. The bound is 0.73339. The best linear forecaster attains it; a rounded, a clipped, a biased and a nonlinearly shrunk forecaster give 0.585, 0.635, 0.676 and 0.732 (Monte Carlo, $3\sigma = 0.0024$). A forecaster holding a second independent record reaches 0.884. The bound is relative to the record, as it must be: more record is a different task.

2.3 The second term

Theorem 2.3 (Second-order frontier law)**.** [FD] *Let the logarithmic growth be $S_n = \sum_{i \le n} X_i$ with independent increments of mean $\lambda > 0$, variance D and finite third moment, and let $q > \frac{1}{2}$. With $L = \ln(\varepsilon/\sigma_0)$, the endpoint horizon is*

$$n_q(L) = \frac{L}{\lambda} - z_q \sqrt{\frac{DL}{\lambda^3}} + O(1). \tag{6}$$

If $\ln |(U^n)'| - n\lambda$ is a coboundary, $g \circ U^n - g$, then $D = 0$ and the correction is bounded in law.

Proof. By the Berry–Esseen theorem $\Pr(S_n \le L) = \Phi((L - \lambda n)/\sqrt{Dn}) + O(n^{-1/2})$. The condition is $\lambda n + z_q \sqrt{Dn} = L$ up to a bounded shift in n; solve for n. The coboundary statement is the definition. □

Exact binomial computation for increments ± 1 with probability $\frac{3}{4}$ gives the scaled correction -2.546, -3.041, -3.111, -3.129 at $L = 50$, 800, 12800, 204800; the predicted limit is -3.139. A second cocycle, the skew tent map with slopes 4 and $\frac{4}{3}$, gives -1.449 against -1.446. For the logistic map at $r = 4$, with $x = \sin^2 \pi\theta$,

$$\ln |(f^n)'(x)| = n \ln 2 + \ln |\sin(2\pi 2^n \theta)| - \ln |\sin(2\pi\theta)|, \tag{7}$$

and the variance of the fluctuation is 1.643, 1.645, 1.645 at $n = 10, 20, 30$, the constant $\pi^2/6$. For deterministic maps Eq. (6) needs a central limit theorem for the finite-time exponent; that is IMPORTED where it holds and is not claimed in general.

2.4 The hole caps the horizon

Theorem 2.4 (Hole cap and closure certificate)**.** [FC] *Let $x_n = A^n x_0$ on $\mathbb{R}^d$, let the initial record have kernel N, and let the uncertainty of x_0 on N have scale at least η. For a later record $c^\top x_n$:*

1. *the forecast standard deviation is at least $\eta \, \| P_N A^{n\top} c \|$ for every recorded precision σ_0, so the horizon is at most the first time this quantity exceeds ε/z;*
2. *the cap is absent iff $c^\top A^k N = 0$ for $k = 0, \dots, d-1$. Then the hole never reaches the record and the frontier (1) holds with the exponent seen by c.*

Proof. (1) The posterior covariance dominates $\eta^2 P_N$. (2) By the Cayley–Hamilton theorem the powers A^k, $k \ge d$, are combinations of the first d. □

Lemma 2.5 (Fibre)**.** *If two states with the same record have later records $2g$ apart, every forecast errs by at least g for one of them.*

Clause (2) is the descent criterion for the pair (initial record, later record), with a finite certificate. It is Kalman's decomposition [2], IDENTIFIED, and GR Automation used its static form. In the battery a hole of scale 10^{-3} gives horizons $15, 18, 18, 18, 18$ as σ_0 runs from 10^{-2} to 10^{-10}, equal to the cap computed from the hole alone. With a closed hole the horizons are $11, 13, 24, 35, 47$, with slope 2.46 against $1/\lambda = 2.5$. A nilpotent chain shows that d checks are needed: the hole first reaches the record at step $d - 1$.

3 Refinement, signature and the free field

Theorem 3.1 (Refinement law)**.** [FE] *Let* $\sum_i s_i \partial_i^2 u = 0$ *with* p *signs* -1 *and* q *signs* $+1$*, discretized on a cubic lattice of spacing* a*. For evolution along a direction* e*, the fastest mode grows at the rate* $(2/a)\sqrt{m_e}$*, where* m_e *is the number of other directions with the sign of* e*. The best direction gives*

$$\Lambda(a) = \frac{2}{a}\sqrt{\min(p,q)-1} \quad (\min(p,q) \geq 1), \qquad \Lambda(a) = \frac{2}{a}\sqrt{D-1} \quad \textit{(definite)}, \tag{8}$$

and the horizon of a forecast that must hold on all resolved modes is $\ln(\varepsilon/\sigma_0)/\Lambda(a)$*. It is proportional to* a*, and independent of* a *exactly when* $\min(p,q) = 1$*.*

Proof. A mode $e^{\mu x_e + ik\cdot x}$ obeys $\mu^2 = s_e \sum_{i\neq e} s_i \hat{k}_i^2$ with $\hat{k}_i^2 = (4/a^2)\sin^2(k_i a/2) \leq 4/a^2$. The positive terms are those with $s_i = s_e$. □

This is the signature theorem of the Tensor Derivation with its premise made operational. There, a predictive update had to be Lipschitz. Here the statement is about a number an observer can read: each halving of the spacing halves the horizon, unless the form is Lorentzian. The computed rates at $N = 8, 16, 32$ are $16, 32, 64$ for $(2,2)$ and $(2,3)$, and $22.6, 45.3, 90.5$ for $(0,3)$; they vanish for $(1,1)$, $(1,2)$, $(1,3)$ and $(3,1)$.

Theorem 3.2 (Isometries of a conserved form)**.** [FE] *Let* $A^\top H A = H$ *with* H *symmetric and nondegenerate.*

1. *If* H *is definite,* $\|A^n v\|_H = \|v\|_H$ *for all* n*.*
2. *If* $Av = \mu v$ *with* $|\mu| \neq 1$*, then* $v^* H v = 0$*, and* $1/\mu$ *is also an eigenvalue.*

Proof. (1) is the hypothesis. (2) $v^*Hv = (Av)^* H (Av) = |\mu|^2 v^* H v$; and $A^{-1} = H^{-1} A^\top H$ is similar to $A^\top$. □

Corollary 3.3 (A free Lorentzian field has no frontier)**.** *A linear field with a positive conserved energy propagates forecast errors isometrically in the energy norm. Its horizon is zero or infinite, and has no slope in* $\ln(1/\sigma_0)$*.*

A finite precision slope in a Lorentzian realization is therefore a record of something else: of interaction, or of curvature, in which the conserved quadratic form of the deviation is indefinite and the growing directions are null for it (Section 4). The static setting of GR Automation is the frontier-free sector: its hole is invariant under the flow for the same reason. In the battery, a random isometry of a positive form keeps the norm to 3×10^{-15} over 200 steps. A random isometry of a form of signature $(4,2)$ has multipliers 13.76, 4.096, 0.2441, 0.0727, paired to 4×10^{-14}, with null eigenvectors to 6×10^{-16}.

Proposition 3.4 (Interacting field: a horizon bound independent of the spacing)**.** [FF] *For the lattice field* $\ddot{q} = \kappa \Delta_a q - m^2 q - \lambda q^3$ *in three space dimensions, every perturbation obeys*

$$\frac{\mathrm{d}}{\mathrm{d}t}\|(z,\dot{z})\|_{E,a} \leq \Lambda \, \|(z,\dot{z})\|_{E,a}, \qquad \Lambda = 3\lambda \, C_S \max(\kappa^{-1/2}, m^{-1}) \sup_t \|q\|_{6,a}^2, \tag{9}$$

and Λ *is bounded independently of* a *by the energy. The horizon is at least* $\ln(\varepsilon/\sigma_0)/\Lambda$*.*

Proof. The perturbation is forced by $-3\lambda q^2 z$, and $\|q^2 z\|_2 \leq \|q\|_6^2 \|z\|_6 \leq C_S \|q\|_6^2 \|z\|_{H_a^1}$ by Hölder's inequality and the uniform discrete Sobolev bound of the Quantum Gravity print. □

Predictive closure therefore survives the quartic interaction in $3+1$ dimensions, and that power is the endpoint at which this argument works. The battery follows a smooth, an alternating and a white perturbation at $N = 8, 12, 16$ with $\lambda = 3$: the growth rates stay below 0.0025 in absolute value at every spacing. With the elliptic sign the alternating perturbation grows at 27.7, 41.6, 55.4.

4 The tensor

4.1 The slope does not depend on the record

Theorem 4.1 (Degree and coordinate invariance)**.** [TA] *Let ψ be a continuous record of the unstable coordinate near x_0, and let $\delta^*(\varepsilon)$ be the largest radius with $|\psi(x_0+\delta)-\psi(x_0)|\le\varepsilon$ for $|\delta|\le\delta^*$. Then*

$$n_q=\frac{\ln\delta^*(\varepsilon)-\ln\sigma_0-\ln Q_q(|Z|)}{\lambda}. \tag{10}$$

The slope is $1/\lambda$ for every record. If ψ vanishes to order k at x_0, the record error grows at rate $k\lambda$ and the initial error enters to the power k; the two cancel.

Proof. The record leaves its tolerance when the state error $\sigma_0|Z|e^{\lambda n}$ reaches δ^*. □

A stress tensor is a quadratic record. Its error grows at twice the exponent, which is the doubling seen in the Fisher form, and its horizon has the slope of the state. Seven records, three powers at zero background and four smooth maps at a background, give the slope 1.428571 against $1/\lambda=1.428571$. The record changes the intercept only. This is Fisher naturality in the language of horizons.

4.2 Curvature is the generator of the forecast

Let γ be a timelike geodesic with unit tangent u and a parallel orthonormal frame. A neighbouring free carrier, initially at relative rest, has separation $\xi(t)=\Phi(t)\xi_0$ with

$$\ddot\Phi=-E(t)\,\Phi,\qquad \Phi(0)=1,\quad \dot\Phi(0)=0,\qquad E_{ij}=R_{i0j0}. \tag{11}$$

E is the tidal tensor; it is symmetric by the pair symmetry of the Riemann tensor of the Levi-Civita connection.

Theorem 4.2 (Tidal frontier)**.** [TB, TC]

1. *$E(0)=-\ddot\Phi(0)$: the tidal tensor is the second time derivative of the forecast propagator at coincidence.*
2. *If E is constant in the frame, the forecast of the separation along an eigendirection with eigenvalue $e_i<0$ has the frontier (1) with exponent $\lambda_i=\sqrt{-e_i}$; along $e_i>0$ the separation oscillates with $\omega_i=\sqrt{e_i}$ and has no frontier.*
3. *$\operatorname{tr}E=\operatorname{Ric}(u,u)$. In vacuum with cosmological constant, $\sum_i e_i=-\Lambda$. For $\Lambda=0$ and $E\neq0$ there is at least one unstable direction, and*

$$\sum_{unstable}\lambda_i^2=\sum_{stable}\omega_i^2. \tag{12}$$

Proof. (1) and (2) are Eq. (11). (3) is the contraction $R_{i0i0}=R_{00}$ and the vacuum equation. □

The conserved form of the deviation is $\frac12|\dot\xi|^2+\frac12\xi^\top E\xi$. It is indefinite when E has a negative eigenvalue, and the growing solution $\dot\xi=\lambda\xi$ is null for it, as Theorem 3.2 requires.

Geometry	E in the frame	Frontier
Schwarzschild	$\frac{M}{r^3}\operatorname{diag}(-2,1,1)$	radial, local exponent $\sqrt{2M/r^3}$; $\lambda_r^2=2\omega_t^2$
Schwarzschild–de Sitter	trace $-\Lambda$	as above, shifted by $-\Lambda/3$

Geometry	E in the frame	Frontier
de Sitter	$-H^2\,\mathbf{1}$	three directions, exponent H
Friedmann, matter era	$+\frac{2}{9t^2}\mathbf{1}$	none: focusing in every direction
Newtonian potential $-M/r$	$\frac{M}{r^3}\,\mathrm{diag}(-2,1,1)$	the same tensor: $E_{ij}=\partial_i\partial_j\phi$

All rows are computed symbolically. For Schwarzschild the components are those of radial free fall, and a pair of observers falling from rest at infinity reproduces $2M/r^3$ to 2×10^{-4}. The last row is the quadrupole $(2,-1,-1)$ of the Tides print: the ocean tide and the tidal frontier are one tensor, and in the Newtonian limit the vacuum trace condition is Laplace's equation. In the Λ era, with $H_\Lambda = H_0\sqrt{\Omega_\Lambda}$ and the Planck values [24] used in Observer Bounds, $1/H_\Lambda = 17.5$ Gyr: one decade of initial precision in the relative position of two unbound carriers buys $\ln 10/H_\Lambda = 40.4$ Gyr of forecast.

4.3 The source is the rate of the Fisher volume

Let the later positions be recorded with noise covariance R. The Fisher form of the initial separation is $I_t = \Phi(t)^\top R^{-1}\Phi(t)$.

Theorem 4.3 (Volume law)**.** [TC] *For every R,*

$$\frac{\partial^2}{\partial t^2}\ln\det I_t\Big|_{t=0} = -2\ \mathrm{tr}\,E(0) = -2\,\mathrm{Ric}(u,u). \tag{13}$$

Proof. $\Phi(t) = 1 - \frac{1}{2}E(0)t^2 + O(t^3)$, so $\det\Phi = 1 - \frac{1}{2}t^2\,\mathrm{tr}\,E(0) + O(t^3)$ and $\ln\det I_t = 2\ln\det\Phi - \ln\det R$. □

Where $\mathrm{Ric}(u,u) > 0$ the Fisher volume falls: the later positions become easier to forecast and the initial separation harder to infer. In vacuum the volume is stationary to second order and only the shape changes. This is the split of the Riemann tensor read as information: the Ricci part moves the volume of the forecast ellipsoid, the Weyl part shears it. On 40 random tidal tensors Eq. (13) holds to 2×10^{-5} by finite differences; in the vacuum cases the axis ratio of $\Phi(\frac{1}{2})$ is at least 1.16. *Canary:* the propagator on the full phase space of separation and relative velocity has determinant one for every E (Liouville). The volume law is a statement about the position record with comoving preparation, not about phase space.

Theorem 4.4 (The response law is the volume law for all observers)**.** [TD] *Let $T_{\mu\nu}$ be the source of the Tensor Derivation, $\rho_u = T(u,u)$ and S_u the trace of its spatial part for the observer u.*

1. *If $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, then for every free observer*

$$\frac{\partial^2}{\partial t^2}\ln\det I_t\Big|_0 = -\kappa\,(\rho_u + S_u) + 2\Lambda. \tag{14}$$

2. *Conversely, if (14) holds at a point for all u in an open set of unit timelike vectors, the response law holds at that point.*

3. *A family of observers whose velocities span, with the time direction, a subspace of dimension m determines exactly $m(m+1)/2$ components of the source: 1, 3, 6, 10 for one observer, collinear, coplanar and generic velocities. The remaining components are in the hole of that family.*

Proof. (1) Trace reversal gives $\mathrm{Ric}(u,u) = \kappa(T(u,u) + \frac{1}{2}T) - \Lambda$ and $T(u,u) + \frac{1}{2}T = \frac{1}{2}(\rho_u + S_u)$; insert in (13). (2) The symmetric tensor $X = \mathrm{Ric} - \kappa(T - \frac{1}{2}Tg) - \Lambda g$ satisfies $X(u,u) = 0$ on an open set of the unit hyperboloid, hence on an open cone by homogeneity, and a quadratic form that vanishes on an open set vanishes. (3) The data are the values of a quadratic form on a subspace. □

Clauses (1) and (2) are Baez and Bunn's formulation of Einstein's equation [3], IDENTIFIED; what is added is the Fisher reading and clause (3). For the programme this means three things.

- The response law, derived in the Tensor Derivation from relabeling, has a form that refers only to records: a forecast, its Fisher volume, and the source. It can be tested without the derivation.
- The ten components of the General Tensor, ρ, q_i, p, π_{ij}, are not seen by one observer. One observer's volume gives $\rho + 3p$. Flux enters only for moving observers, and only along their velocities. In the battery, two sources with equal density and opposite flux agree exactly for eight observers moving orthogonally to the flux and differ by 26.8 for one moving along it. This is the quotient obstruction of the supplied analysis, with its fibre computed.
- Ten generic observers recover a random source to 5×10^{-15}; nine leave rank 9. *Canary:* the law without trace reversal predicts twice the focusing for dust.

5 The light ring: a tensor frontier with a published number

Proposition 5.1 (Photon orbits). [TE] *In the Schwarzschild geometry the photon orbit equation is $u'' + u = 3Mu^2$, $u = 1/r$, in the azimuth φ. The circular orbit $u = 1/3M$ has impact parameter $b_c = 3\sqrt{3}M$ and is unstable with exponent one per radian, that is $\lambda_c = 1/(3\sqrt{3}M)$ in time. For $b = b_c(1+\delta)$ the total azimuth is $\ln(1/\delta) + C + o(1)$.*

Proof. Linearize at the circular orbit: $\delta u'' = \delta u$. The logarithmic divergence is Darwin's [5]. □

This is Eq. (1) for a null record: to forecast a photon through n further half orbits, its impact parameter must be known $e^{-\pi n}$ times better, 1.36 decimal digits per half orbit. The battery integrates the nonlinear orbit equation for $\delta = 10^{-3}, \ldots, 10^{-8}$: the slope of the azimuth in $\ln(1/\delta)$ is 0.9998, and one decade of δ adds 2.30258 radians.

The same exponent is heard. In the eikonal limit the quasinormal frequencies of a black hole are $\omega \simeq (l + \frac{1}{2})\Omega_c - i(n + \frac{1}{2})\lambda_c$, with Ω_c the angular frequency of the light ring and λ_c its instability exponent [6, 7]; IDENTIFIED. The battery computes the Schwarzschild gravitational frequencies by Leaver's continued fraction [8].

l	2	3	4	8	12
$M\omega_R$	0.37367	0.59944	0.80918	1.60619	2.38554
$-M\omega_I$	0.08896	0.09270	0.09416	0.09567	0.09597
$-\omega_I/(\lambda_c/2)$	0.9245	0.9634	0.9786	0.9942	0.9974

The defect falls like $(l + \frac{1}{2})^{-2}$, with coefficient 0.42 to 0.41 between $l = 8$ and $l = 12$. The ringdown decay time is twice the forecast e-folding time of the light ring, exactly in the limit and to 8% at $l = 2$.

For the remnant of GW150914, of detector-frame mass $68\,M_\odot$ and spin 0.67 [10], three numbers can be set side by side.

- The Kerr light ring, computed here from the geodesic potential, has $M\lambda_c = 0.17651$. Twice its e-folding time is 3.80 ms.

- An IMPORTED fit to the Kerr $l = m = 2$ mode [9] gives 248.4 Hz and 4.06 ms. At zero spin the fit agrees with the Leaver values computed here to 1.4% and 0.9%.
- The published values are 251 ± 8 Hz and 4.0 ± 0.3 ms at 90% confidence [10].

The status must be stated exactly. The published pair is the prediction of general relativity for the least-damped mode from the mass and spin inferred over the whole signal; the damped-sinusoid fit to the data after merger is consistent with it from 3 ms after the peak [10]. It is a consistency of records, not an independent measurement of λ_c. The decay time alone does not see the spin (3.76 ms at zero spin); the frequency does (177.6 Hz at zero spin).

Prediction 1 (Photon subrings)**.** Successive images of an emitter around a Schwarzschild black hole are demagnified by $e^{-\pi} = 0.0432$ per half orbit, and by $e^{-\gamma}$ with the Kerr exponent for a rotating one. This is the frontier of Proposition 5.1. It is IDENTIFIED with the photon-ring prediction [11] and awaits interferometric baselines long enough to resolve the first subring.

6 Sources are moments

A forecast law is a statement about probability. A source is an expectation of a quadratic record. The first does not give the second.

Theorem 6.1 (Transfer inequality)**.** [SA] *If a forecast error δ has coverage q at tolerance ε and fourth moment m_4, then*

$$\mathbb{E}\,\delta^2 \le \varepsilon^2 + \sqrt{(1-q)\,m_4}. \tag{15}$$

The bound is attained. Without a moment bound nothing holds: errors of amplitude $1/a$ with probability a^2 have coverage $1 - a^2 \to 1$ and mean square one.

Proof. Split at $|\delta| = \varepsilon$ and apply the Cauchy–Schwarz inequality to the tail. For the spike family $\varepsilon = 0$, $1 - q = a^2$ and $m_4 = a^{-2}$. □

Gate R2 of the Tensor Derivation asks that sources converge as first variations and names one obstruction, oscillation: $\sin(nx)/n$ and the alternating lattice field. The spike family is the other one, concentration. Theorem 6.1 is the budget that excludes it: a frontier certifies a source only together with a fourth-moment bound, that is with uniform integrability. The inequality holds with slack on 2000 seeded discrete laws.

Theorem 6.2 (Forecast deficit)**.** [SA] *Let the state have forecast $\hat{x}_n = A^n \hat{x}_0$ and covariance $\Sigma_n = A^n \Sigma_0 A^{n\top}$, and let $T(x) = x^\top M x$ with $M \succeq 0$. Then*

$$\mathbb{E}\,T(x_n) = T(\hat{x}_n) + \operatorname{tr}(M\Sigma_n) \;\ge\; T(\hat{x}_n). \tag{16}$$

The defect grows at twice the exponent. If A is an isometry of M, it is constant.

Proof. Expand the quadratic form; $\operatorname{tr}(MA^n\Sigma_0 A^{n\top}) = \operatorname{tr}(A^{n\top} M A^n \Sigma_0)$. □

This is the energy deficit of GR Automation, moved from the static hole to the forecast. There, the least-energy field compatible with the readings underestimates the energy by the energy of the hole, and the deficit has one sign. Here, the source of the forecast underestimates the expected source by the stress of the unresolved covariance, with the same sign, and the missing part grows. In the battery the identity is exact, the growth rate is 1.2000 for $\lambda = 0.6$, the defect is constant to all digits under an energy isometry, and the horizon of the source at an absolute tolerance has slope 1.66 against $1/\lambda = 1.67$, as Theorem 4.1 requires.

7 The frontier is the observer's boundary

The frontier of Sections 2 to 4 is a limit in precision. A second limit does not depend on precision at all: an observer's record covers a bounded region, and what lies outside reaches the forecast at a finite speed. This section shows that the two are one object, and says exactly what it means that this speed is the speed of light.

7.1 The causal cap

Theorem 7.1 (Causal cap)**.** [CA] *Let the update have an exact cone of speed C: the state at a point after time n depends only on initial data within distance Cn. Let the initial record cover the ball of radius R about x_0 with precision σ_0, and let the exterior be unknown at scale η. For a later record at x_0:*

1. *for $Cn \le R$ the exterior does not enter. The certificate of Theorem 2.4 holds exactly up to the crossing time R/C;*
2. *for $Cn > R$ the forecast standard deviation is at least η times the leakage of the exterior, for every σ_0;*
3. *for a massless carrier, whose front carries an amplitude of order one, the horizon as $\sigma_0 \to 0$ is exactly R/C whenever $z\eta > \varepsilon$. Hence*

$$H(R) = \frac{R}{C}, \qquad C = \frac{\mathrm{d}R}{\mathrm{d}H}. \tag{17}$$

Proof. (1) is finite propagation, read as $c^{\top} A^k N = 0$ for $Ck \le R$. (2) is Theorem 2.4(1). (3) At the first step beyond the crossing time the leakage is of order one. □

For a field equation the cone is fixed by the principal form alone: a mass term, a potential or the cubic interaction of Proposition 3.4 does not move the domain of dependence [12]. A free Lorentzian field has no frontier in precision (Corollary 3.3); it has this one. In general the horizon is the smaller of the two,

$$n \le \min\Big(\frac{\ln(\varepsilon/\sigma_0) - \ln Q_q}{\lambda},\ \frac{R}{C}\Big), \tag{18}$$

and only the first term can be bought with precision. Equation (17) is the round-trip law $M = \text{latency} + 2N/C$ of the supplied estimator, $C = 2/(\mathrm{d}M/\mathrm{d}N)$, with the horizon in place of the echo time. The round-trip form is the right one: it uses one clock and needs no convention of simultaneity.

The battery uses a split-step walk with an exact cone of one site per step. The leakage of the exterior is exactly zero for $n \le R$, for a massless and for a massive carrier, and equals $\cos^R \theta$ at $n = R+1$ to 2×10^{-16}. For the massless carrier the cap equals R at every R from 20 to 200 and every ratio $z\eta/\varepsilon$ from 2 to 10^9, and the horizon is $30, 30, 30$ at $R = 30$ for $\sigma_0 = 10^{-2}, 10^{-5}, 10^{-8}$. *Canary:* a record of the whole lattice has no cap.

7.2 Front, group and threshold

Proposition 7.2 (What a carrier can show)**.** [CB] *In the walk with coin angle θ, which is the mass in lattice units:*

1. *the front moves at the cone speed for every θ, with amplitude $\cos^R \theta$ at distance R;*
2. *the largest group velocity is $\cos\theta$;*

3. *the speed* $\mathrm{d}R/\mathrm{d}H$ *estimated at sensitivity* $\rho = z\eta/\varepsilon$ *equals the cone speed for* $\theta = 0$ *at every* ρ*. For* $\theta > 0$ *it increases with* ρ*, never exceeds the cone speed, and at fixed* ρ *tends to* $\cos\theta$ *for large* R*.*

Proof. (1) Only the path that never turns reaches the edge. (2) The dispersion relation is $\cos\omega = \cos\theta\cos k$, so $\mathrm{d}\omega/\mathrm{d}k = \cos\theta\sin k/\sin\omega$, largest at $k = \pi/2$. (3) is computed. □

sensitivity ρ		1.5	10^3	10^9	10^{30}
$\theta = 0$		1.000	1.000	1.000	1.000
$\theta = 0.3$	$(\cos\theta = 0.955)$	0.922	0.985	1.000	1.000
$\theta = 0.8$	$(\cos\theta = 0.697)$	0.598	0.719	0.777	0.993

Mass and dispersion move the bulk and the threshold arrival. They do not move the front. Two consequences follow for any measurement of a frontier. A dispersion gate must compare fronts, or arrivals at one threshold; and a massive carrier that shows a lower speed at finite sensitivity is below the frontier, not a counterexample to it. The frontier is the supremum over carriers and sensitivities. At finite resolution a massive carrier cannot reach it: its observable speed is $\cos(ma)$ times the cone speed, and tends to the cone speed as the spacing goes to zero.

7.3 Maximal means invariant

Theorem 7.3 (The frontier is the invariant speed)**.** [CC] *Let inertial observers be related by a group of linear transformations that respects isotropy, so that there is a constant* $K \in (0, \infty]$*, the same for all, with velocities composing as* $w' = (w - v)/(1 - wv/K^2)$*.*

1. *The only speeds on which all observers agree are* $\pm K$*.*
2. *If records respect one causal order, so that emission precedes reception for every observer, then every signal speed is at most* K*.*
3. *Hence the frontier, the supremum of signal speeds, is at most* K*; and a carrier whose two-way speed is the same for all observers travels at* K *and attains it.*

Proof. (1) The composition law fixes w for all v only if $w^2 = K^2$. (2) For a signal of speed w the interval between emission and reception transforms as $\Delta t' = \gamma\,\Delta t\,(1 - vw/K^2)$, which is negative for $K^2/w < v < K$ when $w > K$. (3) follows. □

The existence of K from the relativity principle is von Ignatowsky's theorem [13], and clause (2) is the reply-loop argument [14]; both IDENTIFIED. In the battery a reply sent at $4K$ between observers receding at $0.8K$ arrives 0.547 time units before the question was sent, while for $w \le K$ the order is kept at every tested velocity. An observer with Lorentzian standards, a contracted rod and a dilated clock, measures the two-way speed K in sixty random directions at $0.1K$, $0.5K$ and $0.9K$ to 7×10^{-16}. *Canary:* in a medium, with Galilean standards, the two-way speed is $c(1 - \beta^2)$ along the motion and $c\sqrt{1 - \beta^2}$ across it.

Theorem 7.3 fixes what a match to the speed of light can mean. Resolution Geometry has no internal unit, so no number of metres per second can be derived. What the programme derives, under the premises of the Tensor Derivation and Theorem 3.1, is one Lorentzian form, hence one finite K. What data can decide is whether the frontier of actual carriers is that K: whether light's two-way speed is the same for all observers, and whether anything is faster. Since 1983 the metre is defined by this speed. Metrology has adopted the statement that the observer's ruler is the frontier times the observer's clock.

Remark 7.4 (A relation that was tested and rejected). For de Sitter space the frontier exponent H times the radius c/H of the observer's horizon equals c. For the Schwarzschild light ring the exponent λ_c times the shadow radius b_c also equals c. This is not a law: the product is $\frac{1}{2}c$ for the surface gravity and radius of the Schwarzschild horizon, and $0.643\,c$ for the Kerr light ring at spin 0.67. [CC]

8 The raw speed: gates and data

8.1 Audit of the supplied gates

The supplied synthetic programme estimates $C = 2/(\mathrm{d}M/\mathrm{d}N)$ from counts of clock periods M against counts of ruler periods N, and declares a frontier locked when seven gates pass: finite, absolute recovery, cross-cell spread, isotropy, dispersion, linearity and refinement. The battery regenerates it exactly, with its seed and draw order (locked mean 3.7497161154). Four points need correction before the gates are applied to data. [CD]

- *One gate needs the answer.* Absolute recovery compares with the true value. Without it the precursor scenario passes the six remaining gates. It is a locked frontier at $C/(1-p) = 3.7955$, measured 3.7987; a two-window slope gate that rejects the scale drift at 50σ sees it at 2.2σ. This is the correct reading and not a defect of the data: an arrival that is early in proportion to the distance is a faster frontier, and the slower carrier is then below it (Proposition 7.2). Leakage can be told from a frontier only with arrivals resolved by threshold, or with a second carrier.

- *The lock depends on the seed.* On a truly locked frontier the seven gates pass for 106 of 200 seeds. Every failure is the linearity gate, whose threshold of 0.3% is at the noise level of the maximum over twelve cells. A gate calibrated to the noise, the mean signed curvature against its standard error, passes all 200 and rejects the scale drift at 73σ.

- *The covariance check cannot fail.* It compares $(\alpha/\beta)C$ with itself plus injected noise. Covariance under a change of standards is tested only by physically different rulers and clocks (Section 8.2).

- *A medium passes.* For an observer at rest in a medium, a wave of that medium passes all seven gates. They would lock the speed of sound.

The last point decides the design. By Theorem 7.3 an observer's boundary differs from a medium's wave speed in two ways, and each is a gate.

- *Boost.* A second observer in relative motion, with standards moving with it, finds the same raw value. The medium fails: at $0.2\,c_s$ the mean falls from 3.7502 to 3.6367, and isotropy fails as well. A Lorentzian carrier passes.

- *Maximality.* No carrier of any field is faster. A medium fails as soon as one carrier of another field is included.

8.2 Published raw data

An uncalibrated speed is a count of ruler periods per count of clock periods, with ruler and clock realized independently. Between 1967 and 1983 the metre was 1 650 763.73 wavelengths of a krypton-86 line and the second 9 192 631 770 periods of a caesium transition [15]. In that system the 1972 measurement of a methane-stabilized laser [16], frequency 88 376 181.627(50) MHz against caesium and wavelength 3.392 231 376(12) μm against krypton, is a pure number:

$$C_{\mathrm{Kr,Cs}} = 53\,835.1286 \pm 0.0002 \quad \text{krypton wavelengths per caesium period,} \tag{19}$$

with $\lambda\nu = 299\,792\,456.2 \pm 1.1$ m/s. The value adopted in 1983 corresponds to 53 835.1289. [CE]

Gate	Published result	Source
Finite; one value across rulers, clocks and methods	$299\,796 \pm 4$; $299\,792 \pm 3$; $299\,792.5 \pm 0.1$; $299\,792.4562 \pm 0.0011$ km/s. Pulls $+0.89$, -0.15, $+0.42$, -1.69; $\chi^2 = 3.8$ for 4	rotating mirror 1926; cavity resonator 1947; radio interferometer 1958; laser 1972 [17, 16]
Dispersion, raw	microwave against infrared: $(1.5 \pm 3.3) \times 10^{-7}$	same
Dispersion, photon energy	linear term excluded above $10\,E_{\mathrm{Pl}}$	LHAASO, as in the Quantum Gravity print
Isotropy	$(9.2 \pm 10.7) \times 10^{-19}$ over a year of rotation	[18]
Observer velocity	no variation at 2×10^{-13}	[19]
Source velocity	$k < 2 \times 10^{-9}$ in $c' = c + kv$; γ rays from π^0 at $0.99975\,c$ travel at c to 4×10^{-4}	[20, 21]
Maximality, second field	gravitational against electromagnetic carrier: between -3×10^{-15} and $+7 \times 10^{-16}$	[22]
Maximality, fast-light medium	information arrives no earlier than in vacuum although the group velocity exceeds c	[23]
Linearity over distance	not evaluated here on raw data	—

The bounds on the gravitational carrier are reproduced in the battery from the delay of 1.74 ± 0.05 s, a distance of at least 26 Mpc and an emission lag of at most 10 s: -3.09×10^{-15} and $+6.5 \times 10^{-16}$. *Canary:* the speed of sound in air is 1.1×10^{-6} of the frontier and fails maximality at once.

Status of the match. Every gate that could be evaluated on published data is passed at the bound stated. That is consistency, and it is strong: isotropy at 10^{-18}, a second field at 10^{-15}. It is not a proof, since a bound is not an identity, and linearity over distance was not evaluated here. With Theorem 7.3 the content is this: the two-way speed of light is observer-independent to the bounds above, so it is the invariant speed K to those bounds; no carrier is known to be faster, so the frontier is K; and by Theorem 7.1 that frontier is the cap on every forecast made from a bounded record. The frontier is the observer's boundary. Its value in any units is a convention; its raw value against independent standards is a measured pure number.

9 What is closed and what is not

Statement	Status	Basis
Predictive metric is the generating function of the frontier	PROVED	Theorem 2.1
Exponent is the critical discount	PROVED	exponential profiles
Frontier bounds every forecaster of the record	PROVED	Gaussian case; Anderson
Second-order law	PROVED	independent increments; CLT IMPORTED for maps
Hole cap; closure in d steps	PROVED; IDENTIFIED	Kalman

Statement	Status	Basis
Horizon proportional to spacing unless Lorentzian	PROVED	linear lattice fields
Free Lorentzian field has no frontier	PROVED	positive conserved form
Interacting quartic field: uniform bound	PROVED	smooth data; cubic force; $d = 3$
Slope independent of the record	PROVED	Theorem 4.1
$E = -\ddot{\Phi}(0)$; tidal exponents	PROVED	geodesic deviation
Volume law	PROVED	comoving preparation
Response law $\Leftrightarrow$ volume law for all observers	IDENTIFIED	Baez–Bunn
Observer rank $1, 3, 6, 10$	PROVED	Theorem 4.4
Ringdown rate $\to$ half the light-ring exponent	IDENTIFIED; COMPUTED	eikonal; Leaver ladder
GW150914 numbers	CONSISTENT	published prediction; imported fit
Transfer inequality; forecast deficit	PROVED	Theorems 6.1, 6.2
Causal cap $H = R/C$ of a bounded record	PROVED	exact cone; massless carrier
Front fixed by the principal form; bulk and threshold not	PROVED; COMPUTED	walk; Proposition 7.2
Frontier $\leq$ invariant speed; invariant carrier attains it	IDENTIFIED	von Ignatowsky; reply loop
Supplied seven gates suffice	REFUTED	a medium passes; seed dependence; one gate needs the answer
Speed of light is the frontier	CONSISTENT	published bounds 10^{-9} to 10^{-18}; linearity not evaluated
Value of the frontier in units	NOT DERIVABLE	no internal unit; raw number measured
Derivation of the response law from the frontier	NOT CLAIMED	the law is used, not derived
Gate A; nonlinear completion; inhomogeneous stability; gauge sectors	OPEN	unchanged

The supported statement. *The forecast frontier is not an addition to Resolution Geometry. It is the predictive metric of the Tensor Derivation read as a generating function, and its exponent is a quotient invariant. For the tensor it supplies three readings that use records only: curvature is the generator of the forecast of relative position, the source is the rate of its Fisher volume, and the response law is that rate for all observers. The frontier of a quadratic source has the slope of the state, and its expectation carries a positive defect. For a bounded record it is the cone: no precision buys a forecast beyond the crossing time, and the speed in that statement is the invariant speed, which published data identify with the speed of light to the stated bounds. None of this derives the response law, and the value of the frontier in units is not derivable.*

10 Predictions and falsifiers

- *Critical discount.* For a system with exponent λ, predictive metrics built from records are Lipschitz-equivalent to the present metric for $\beta < e^{-\lambda}$ and Hölder with exponent $\ln(1/\beta)/\lambda$ above. A measured exponent that does not follow $\min(1, \ln(1/\beta)/\lambda)$ falsifies Theorem 2.1 for that system's error profile.

- *Second term.* A system with measured exponent variance rate D has the horizon correction $-z_q\sqrt{DL/\lambda^3}$. The logistic map at $r = 4$ is excluded as a test.
- *Vacuum sum rule.* In vacuum the squared tidal exponent equals the sum of the squared tidal frequencies. A gradiometer in vacuum with a nonzero trace, after rotation terms are removed, falsifies it.
- *Observer rank.* A set of observers with coplanar velocities cannot fix more than six components of a source by forecast volumes. A claim of ten from such a set is an error of analysis.
- *Photon subrings.* The demagnification per half orbit is $e^{-\pi}$ for zero spin.
- *Cap.* For a record of radius R of a massless field the horizon is R/C and does not move with the recorded precision. A forecast that holds beyond the crossing time, with the exterior unknown, falsifies Theorem 7.1.
- *Second field.* A multimessenger event in which a gravitational and an electromagnetic front differ by more than the bound above, after emission lags are accounted for, falsifies maximality as stated.
- *Boost.* A raw speed that changes with the observer's velocity at 10^{-13}, or with orientation at 10^{-18}, makes the carrier a wave of a medium and not the frontier.

11 Limits

- Theorem 2.2 is proved for Gaussian records with a posterior variance independent of the record value. For other laws the posterior quantile replaces z, and the statement is not proved here.
- The tidal exponents are exact for a tidal tensor that is constant in the parallel frame, and local otherwise. Along an infalling geodesic in Schwarzschild they change with r.
- The volume law concerns the second derivative at coincidence. The later evolution is governed by the Raychaudhuri equation [4] and is not used.
- Proposition 3.4 bounds the rate; it does not compute a Lyapunov exponent of the field.
- The ringdown comparison uses published values that are themselves predictions conditioned on general relativity, and an imported fit.
- Nonlinear forecast horizons are computed, not enclosed: no computer-assisted bound on a nonlinear orbit is given.
- The causal cap is proved for an exact cone. A lattice field in continuous time has exponentially small tails outside the cone, and the cap then holds up to a term logarithmic in the sensitivity.
- The data of Section 8.2 are published values and bounds, not reanalysed raw records. The four historical determinations are taken from a secondary table [17] except the 1972 one. The distance and emission lag used to reproduce the bound on the gravitational carrier are recalled, not verified; the bound itself is verified.
- One-way speeds are not addressed. All statements concern two-way speeds.

A Battery

The battery `rg_tensor_frontier.py`, with `rg_curv.py`, needs numpy, scipy and sympy, uses seed 20261113 and runs in about eight seconds. It writes `rg_tensor_frontier_results.json`, exits nonzero on any failure, and reruns are byte-identical in the stated environment. The script `rg_tensor_frontier_mutations.py` applies twenty-two one-line falsifications; each is reported as a failure of its group.

Group	Content and canary
FA	two-sided bound on 2000 random profiles; Hölder exponent for the doubling map; inversion of the family; Lipschitz canary
FB	coverage bound; exact linear forecasters; five nonlinear forecasters; second-record canary
FC	saturation at the cap; closed hole with slope $1/\lambda$; nilpotent-chain canary; fibre lemma
FD	exact binomial horizons for two cocycles; coboundary variance of the logistic map; constant-correction canary
FE	growth rates by signature at three spacings; isometries of definite and indefinite forms; $(2, 2)$ canary
FF	three perturbations of the interacting lattice field at three spacings; Grönwall rate; elliptic canary
TA	seven records, one slope; halved-slope canary
TB	tidal tensors of five geometries, symbolic; free-fall pair; Λ-era numbers; focusing canary
TC	volume law on 40 tidal tensors; $E = -\ddot{\Phi}(0)$; frontier slope; Liouville canary
TD	source recovery from observers; rank table; flux hole; trace-reversal canary
TE	photon azimuth; Leaver ladder $l = 2, \ldots, 12$; Kerr light ring; ringdown numbers; zero-spin canary
SA	transfer inequality on 2000 laws; spike family; deficit identity, growth and isometry
CA	exact closure before the crossing time; cap $= R$ for the massless carrier at every sensitivity and precision; whole-lattice canary
CB	speed by sensitivity for three carriers; bulk below the largest group velocity; front at extreme sensitivity; bulk-arrival canary
CC	invariant speed; causal order; reply loop; two-way speed for Lorentzian standards and for a medium; rejected product rule
CD	supplied programme regenerated; precursor without the true value; 200 seeds; medium at rest and moving; boost and maximality gates
CE	krypton–caesium raw number; pulls of four determinations; microwave against infrared; gravitational-carrier bounds reproduced; sound canary

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